

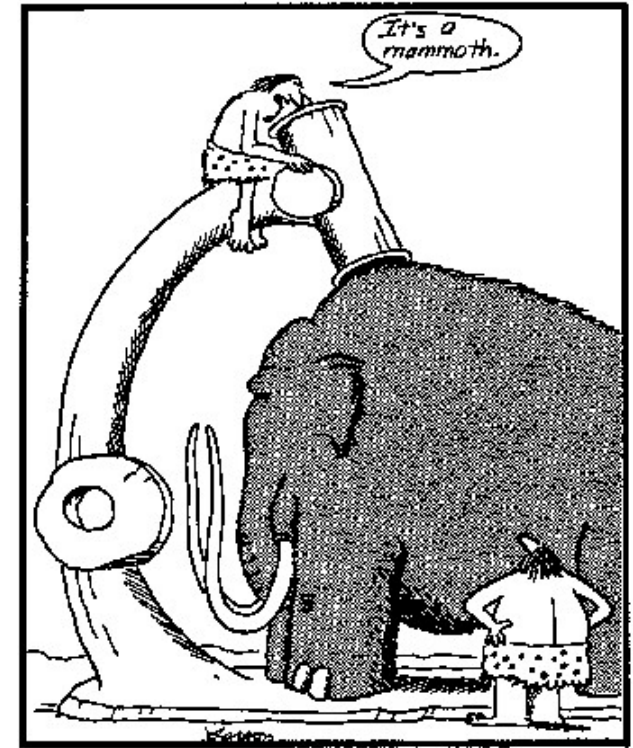
Intro to Microscopy

Mr. Queenan – Biology 9

Unit 1

Goals

- Understand the location and purpose of:
 - Illumination source
 - Lenses
 - Apertures
 - Detectors
- Understand the relationship between size and scale
- Understand what magnification and resolution mean and how they are related



Early microscope

The Microscope

- Compound Light Microscope
 - Most common microscopy tool used in science labs
 - What are other types of microscopes?

Microscope Parts

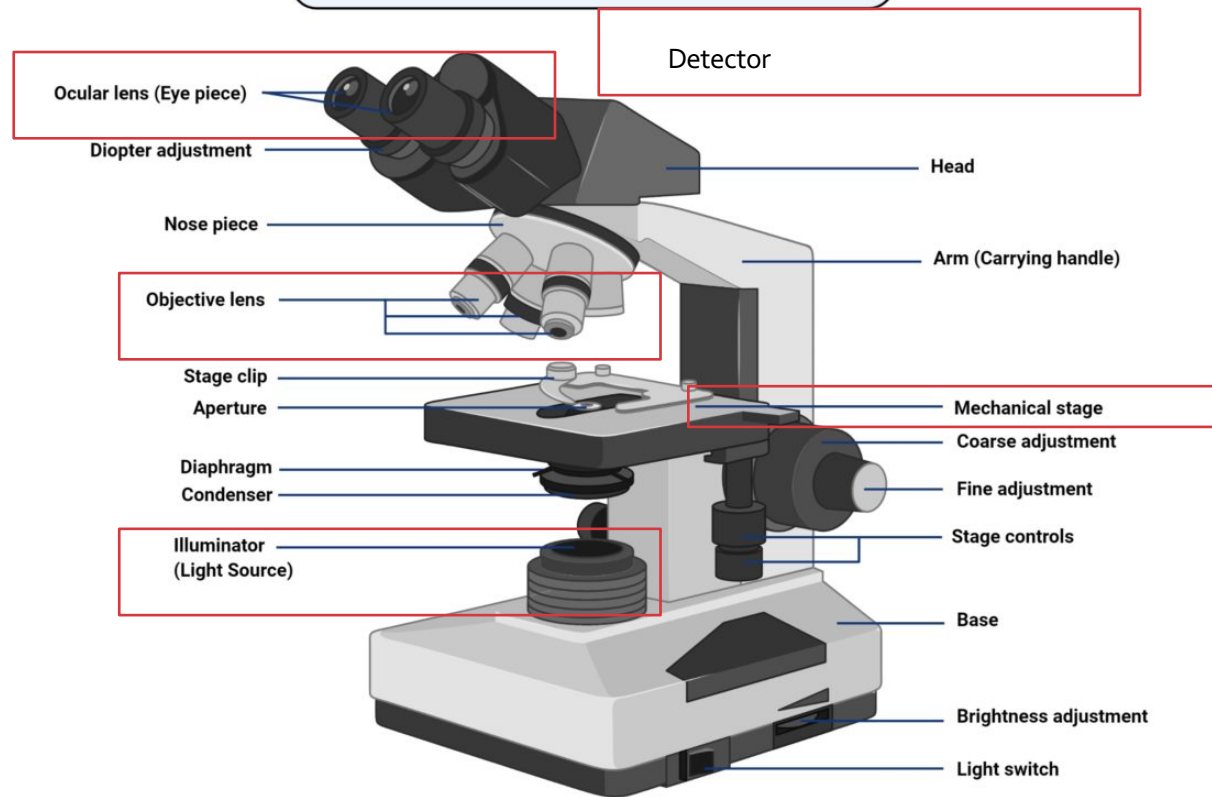


Figure: Parts of a microscope, Image Copyright © Sagar Aryal, www.microbenotes.com

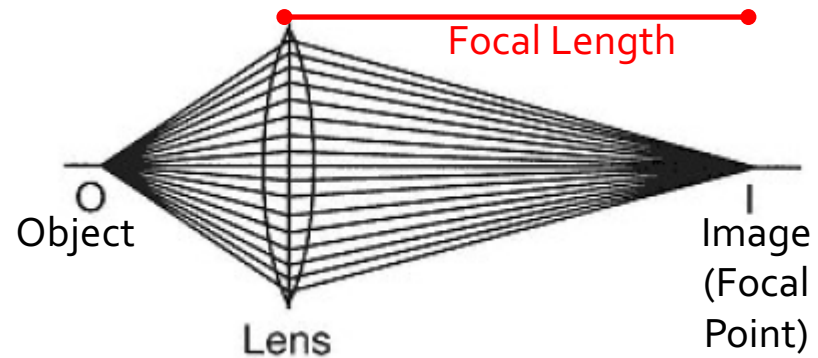
The Microscope

- **Illumination Source**
- The type and source of electromagnetic radiation that is used to illuminate the sample
- Light Microscope (LM)
 - White light generated by LED bulb (tungsten filament light bulb in older microscopes)
- Electron Microscopes (EM)
 - Use beam of focused electrons, usually from a crystal or filament



The Microscope

- **Lens**
 - An optical device with near-perfect symmetry that is able to transmit and focus/bend electromagnetic radiation
 - Shape of the lens determines the direction the radiation will bend
 - Microscopy: Rays (waves) of radiation pass through a lens and converge (come together) at a specific distance from the lens
 - Known as focal length

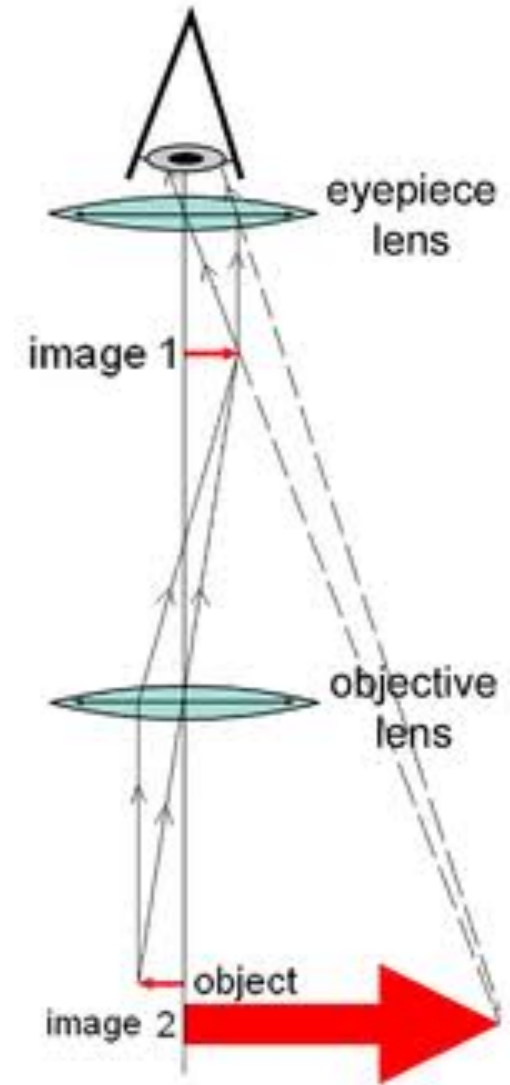


The Microscope

- **Lens**
- Focal length is fixed (unchangeable)
 - To focus the image, the distance from the lens to the object is changed
 - To magnify the image, the lens is replaced with one having a different focal length
- Different lenses have different purposes
 - Condenser – Gathers ER from source onto specimen
 - Objective(s)– Gathers ER from sample to and focuses to detector; can be multiple
 - Intermediate and Projector – Not always found. Used to increase magnification
- Called compound microscope because multiple lenses compound, or multiply the magnifying power of the unit.

The Microscope

- Lens Compounding Practice
 - Ocular Lens = 10x mag
 - Objective Lens – 20x mag
 - Total magnification power =
 - 200x magnification
- Ocular Lens = 10x mag
- Objective Lens – 10x mag
- Total magnification power =
 - 100x magnification
- Ocular Lens = 10x mag
- Objective Lens – 40x mag
- Total magnification power =
 - 400x magnification



The Microscope

- **Stage and Specimen**

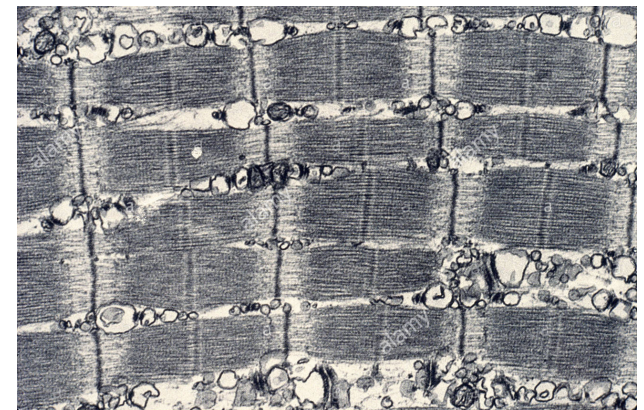
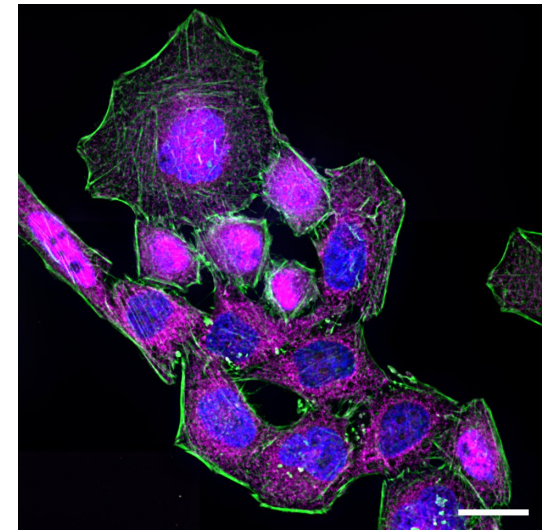
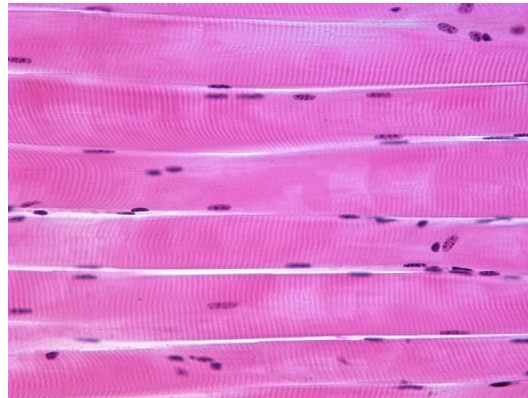
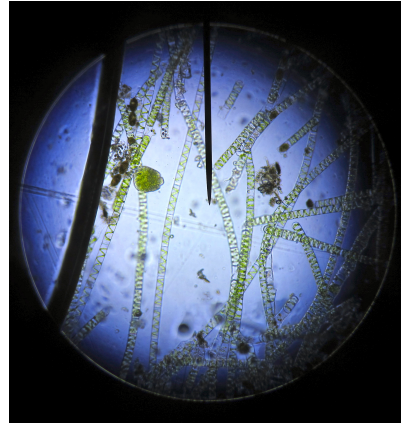
- Thick sections/slices of sample ($\sim 1\mu\text{m}$) or sample small enough for light to pass through
- Samples typically stained with dyes to enhance features
- Mounted on glass slide with coverslip
- Stage has opening to allow light to pass through



The Microscope

- **Detectors & Imaging**

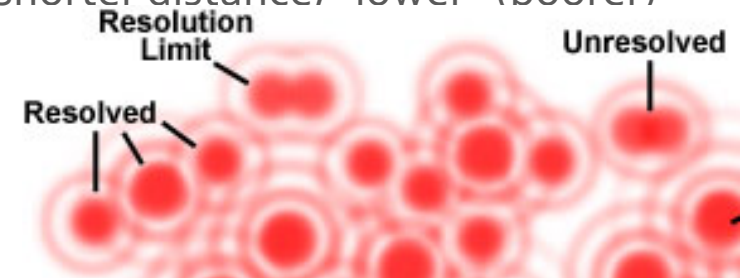
- Device that collects electromagnetic radiation after it has passed through the sample and all of the lenses and displays the final image.
- LM:
 - Eye
 - Digital camera & Computers



Resolution v. Magnification

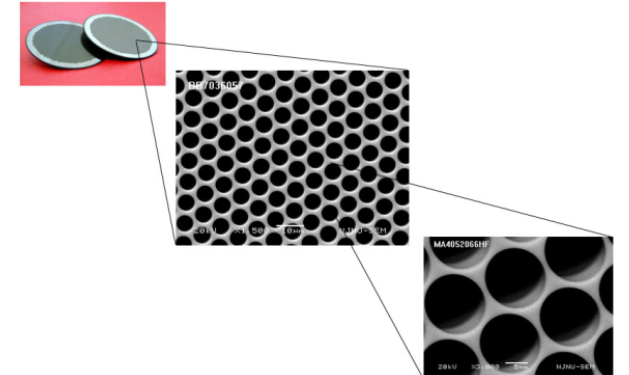
- **Resolution:**

- The ability to distinguish two points which are closely adjacent and would otherwise appear to be one.
- “Higher” (better) resolution is a shorter distance: “lower” (poorer) resolution is a longer distance

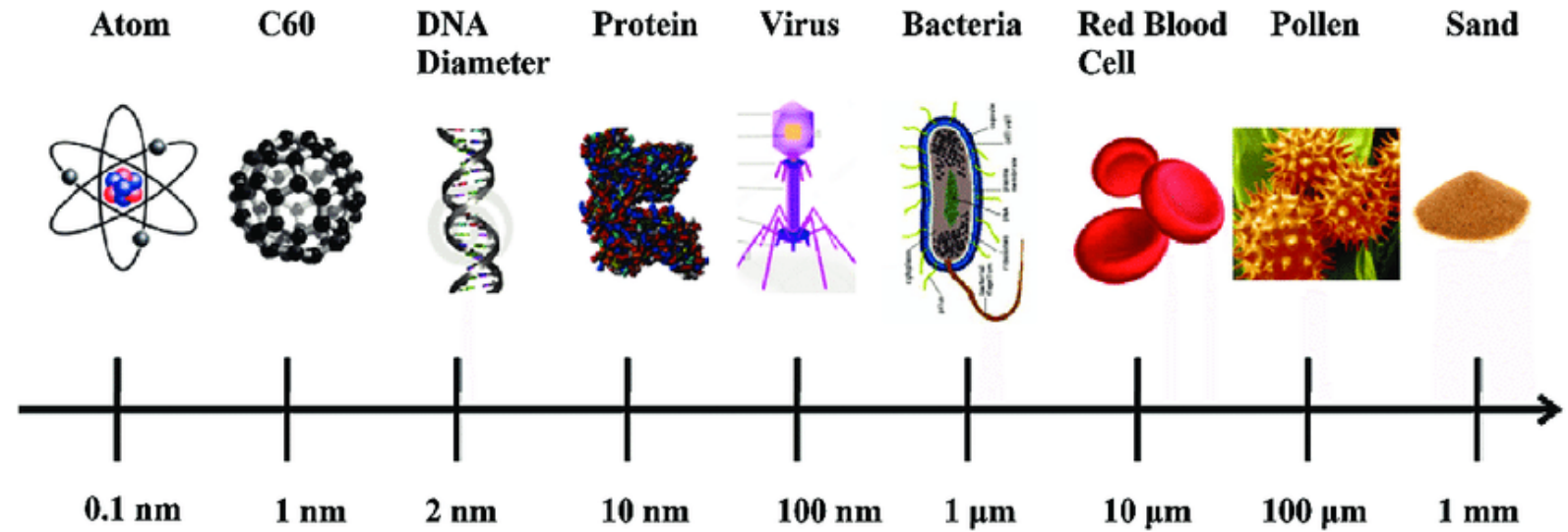


- **Magnification:**

- The degree of enlargement of an image as compared to the original.
- A measure of the increase in diameter of a structure.
- Written as a quantity X (ex: 1000X) for how many times the image is magnified



Size and Scale



Pico < Nano < Micro < Milli < Meter
 10^{-10} - 10^{-12} 10^{-9} - 10^{-7} 10^{-6} - 10^{-4} 10^{-3} >1

Making note of these units and how many decimal places are changed when converting units is important to understand, and will be something you see throughout your science education and careers.

Resolution

- Unaided eye can distinguish two points that are 0.2mm (200 μ m) apart.
 - Any closer together and points will appear as a single object
 - Ex: printed image



- Image is made up of dots too small for unaided eye to see, they appear as one smooth, continuous image until it is magnified
- *Resolution of light microscopy is limited by wavelength, which is why tools like EM are needed. **Smallest resolution is .2 μ m***

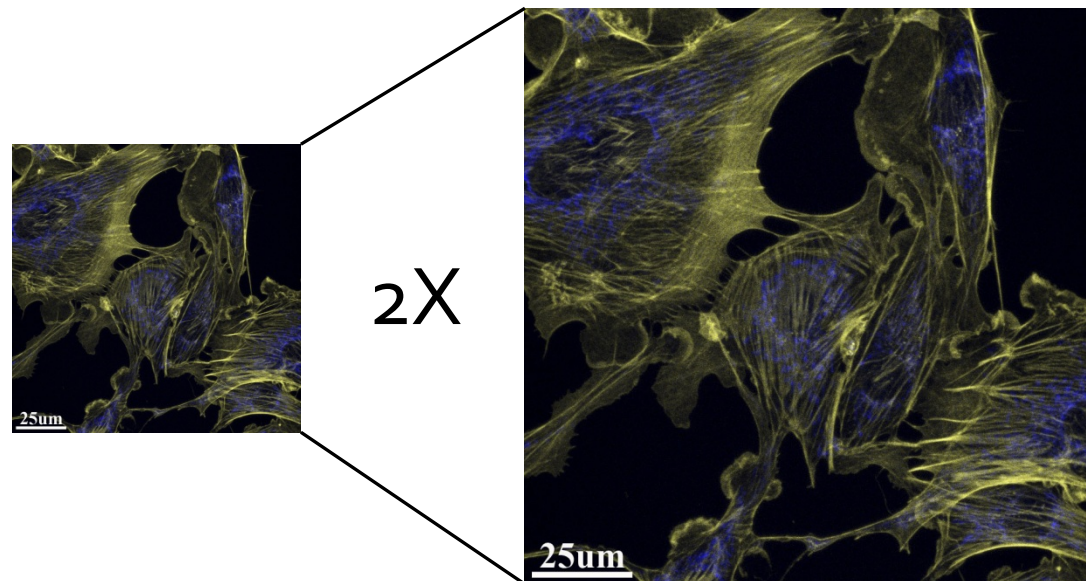
Magnification

- Magnification is equal to the distance measured between 2 points on an image divided by the distance between the same 2 points on the original object.
 - $\text{Mag} = \frac{\text{image distance}}{\text{object distance}}$
- Ex: Calculate the magnification of an image where the diameter of the object is **5 μm** and the diameter on the image is **12.5 mm**.
- Answer:
 - $\text{Magnification} = 12500\mu\text{m}/5\mu\text{m} =$
 - 2500X



Magnification

- **Reliability of Magnification Value**
- The magnification value given by a microscope is only valid if the size of the image is UNCHANGED.
 - Enlarging the image increases the magnification.
 - Shrinking the image reduces the magnification.
- Quality images always have a scale bar; scale bar remains valid when the size of the image is changed.



SEM

- Scanning Electron Microscope
 - Uses electrons rather than visible light as illumination source
 - Wavelength of electrons based on energy sent through the filament
 - Much smaller wavelength than visible light
 - Allows greater resolution than light microscopy
 - SEM scans only the surface of the object of interest
 - Electrons do not pass through the sample – cannot see inside
 - TEM used for this purpose
 - Images only appear in black and white
 - Why?
 - Color is based upon visible spectrum. Electron wavelength outside of that spectrum

